

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network
Security COURSE CODE: CSA5117

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

V.RAKSHITHA

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security

A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

- a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
- b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message

A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

- a. Deduce the most likely original message.
- b. Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System

A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

- a. Demonstrate how the encryption process transforms the message step-by-step.
- b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation

During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

- a. Analyze how investigators could detect the presence of hidden data within the image.
- b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication

A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEFEAOCAIVDEN".

- a. Determine the most probable original message by reversing the encryption process.
- b. Critically evaluate why such a method is vulnerable in modern communication systems.
- c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

1) Scenario: Event communication security

a) Encrypted message using a shift of 4 positions

Given message: MEET AT NOON

Using the caesar cipher, shift each letter 4 positions forward in the alphabet

Original letter	shifted letter
M	Q
E	I
E	I
T	X
A	E
T	X
N	R
O	S
O	S
N	R

Encrypted message: QIIX EX RSSR

b) How to retrieve the original message

An authorized member who knows the encryption rule can decrypt the message by shifting each letter 4 position backward in the alphabet.

For example:

O $\rightarrow$ M

I $\rightarrow$ E

X $\rightarrow$ T

E $\rightarrow$ A

R $\rightarrow$ N

S $\rightarrow$ Q

After shifting all letters 4 positions backward and keeping the spaces unchanged, the original message is recovered.

Decrypted message: MEET AT NOON

Conclusion:- The caesar cipher provides a simple way to encode messages using a fixed shift. However, it is not secure enough for highly confidential information because the encryption can be easily broken.

4) SUSPICIOUS Digital Asset Investigation

a) How can investigators detect hidden data within an image?

Investigators can use the following techniques to detect hidden confidential information in a company logo image:

1) Metadata analysis:-

Examine the image's metadata (EXIF, comments, and file properties) for suspicious information.

2) Steganalysis:-

Use specialized forensic tools to identify data concealed inside the image pixels.

3) File size analysis:-

Compare the image's file size with similar images to identify unusual differences.

4) Visual analysis:-

Adjust brightness, contrast, and colour channels to reveal unusual patterns.

5) Statistical analysis:-

Analyze pixel values and colour distributions to detect irregularities caused by hidden data.

6) File signature analysis:-

check for unexpected data appended to the image file or mismatches between its file extension and actual format.

b) Recommend two techniques to strengthen the security of hidden communications

1) Encryption:-

Encrypt confidential information before embedding it in an image. Even if investigators extract the hidden data, they cannot easily understand it without the decryption key.

2) Steganography:-

Embed encrypted information within image pixels using a suitable method, such as least significant bit (LSB) embedding.

Careful embedding can reduce visible changes, although it does not guarantee that the hidden data will avoid detection.

Conclusion:-

Metadata analysis, steganalysis, and statistical examination help investigators detect hidden information. Encryption and steganography can improve the confidentiality of communications.